

```
import pygame
```

```
import math
```

```
class Robot:
```

```
    def __init__(self, x, y):
```

```
        # World position
```

```
        self.x = x
```

```
        self.y = y
```

```
        # Direction angle
```

```
        self.angle = 0
```

```
        # Visual size
```

```
        self.width = 60
```

```
        self.height = 45
```

```
        # Movement (temporary visualization)
```

```
        self.speed = 3
```

```
        # Sensor visualization
```

```
self.sensor_range = 100
self.front_distance=100
self.left_distance=100
self.right_distance=100
self.rear_distance=100
```

```
def move(self, keys):
```

```
    if keys[pygame.K_UP]:
        self.y -= self.speed
        self.angle = -90
```

```
    if keys[pygame.K_DOWN]:
        self.y += self.speed
        self.angle = 90
```

```
    if keys[pygame.K_LEFT]:
        self.x -= self.speed
        self.angle = 180
```